

Precept 3 Outline

ECO 202 TAs

September 24, 2026

1 Exam Prep (if read before exam)

1. Existing Practice Material:

- (a) Each lecture note contains short practice questions and checkpoints
- (b) Each lecture also has practice problems associated with the lecture, with solutions
- (c) For each precept we post notes with some additional problems not covered in precept, with solutions
- (d) Two practice exams posted with solutions on the github!

2. Other Logistics:

- No calculators allowed on exam!
- No coding/AI problems on exam
- Material up to and including Monday, September 21 lecture is fair game

2 Precept 3 (Lecture 4, 5) Core Concepts

2.1 Lecture 4

1. Correlation, covariance, and relationship between them
2. Least squares minimization problem
 - How they lead to formulas for b_1, b_0
 - Residuals and properties of residuals (Note: can talk through what residuals look like on graph)
3. Properties of Least squares
 - Fitted line passes through $\bar{x}, \bar{y}$
4. R^2 and r (R squared versus correlation, and relationship between them)

2.2 Lecture 5

1. Potential outcome notation: D_i , $Y_i(1)$, and $Y_i(0)$
2. Average Treatment Effect (ATE)

3 Precept practice problems

1. Let y_i , $i = 1, \dots, 265$, be the fraction of students in school district i who qualify for a subsidized lunch, and let x_i be average household income in that school district, measured in thousands of dollars. Suppose

$$\bar{x} = 38.0, \quad s_x = 14.0, \quad \bar{y} = 0.26, \quad s_y = 0.18, \quad r = -0.73.$$

- (a) Given these data, what is your best guess of the fraction of students who qualify for subsidized lunch in a district with average household income equal to \$24,000 (so $x = 24.0$)? Explain.

Solution. For the regression of y on x ,

$$b_{y|x} = r \frac{s_y}{s_x} = -0.73 \frac{0.18}{14} \approx -0.009386,$$

and

$$a_{y|x} = \bar{y} - b_{y|x} \bar{x} \approx 0.6167.$$

Hence

$$\hat{y} = 0.6167 - 0.009386(24) \approx 0.3914.$$

The best linear prediction is therefore 0.391, or about 39.1%.

- (b) Given these data, what is your best guess of average household income in a school district where 0.45 = 45% of students qualify for subsidized lunch? Explain.

Solution. Now the desired response is x , so we must regress x on y :

$$b_{x|y} = r \frac{s_x}{s_y} = -0.73 \frac{14}{0.18} \approx -56.7778,$$

$$a_{x|y} = \bar{x} - b_{x|y} \bar{y} \approx 52.7622.$$

Therefore

$$\hat{x} = 52.7622 - 56.7778(0.45) \approx 27.212.$$

The best linear prediction is \$27,212 approximately. Notice that this is not obtained by algebraically inverting the line from part (a): the regression of x on y and the regression of y on x solve different least-squares problems.

- (c) Why is it problematic to use linear regression here to predict the fraction of students receiving subsidized lunches in very wealthy school districts?

Solution. For sufficiently large x , the fitted line produces $\hat{y} < 0$. A fraction cannot be negative. Very wealthy districts may also lie outside the observed range, creating an extrapolation problem. Thus the linear model is not credible for arbitrarily large incomes.

2. In an effort to understand the effects of the Atkins diet, a randomized experiment is run with 320 initially overweight mice. All mice receive food with the same total number of calories, but the calorie intake is achieved with five different levels of carbohydrates. Let x denote carbohydrate intake in grams per day. After 28 days, the mice are weighed; let y denote weight in grams. We know that

$$\bar{x} = 16.2, \quad \bar{y} = 167.3, \quad s_x = 6.5, \quad s_y = 35.8,$$

and the slope of the least-squares regression line is $b = 4.6$.

- (a) Why can one be reasonably confident that this evidence points toward a causal relationship between diet and weight in mice? Compute a relevant measure of fit and explain why a causal interpretation is appropriate using relevant terminology.

Solution. First recover the correlation from $b = r(s_y/s_x)$:

$$r = b \frac{s_x}{s_y} = 4.6 \frac{6.5}{35.8} \approx 0.835, \quad r^2 \approx 0.698.$$

Thus about 69.8% of the sample variation in mouse weight is accounted for by the fitted linear relationship with carbohydrate intake.

The causal argument comes from the *randomized design*, not from the large value of r^2 . Random assignment makes the treatment groups comparable on average before treatment and breaks systematic association between carbohydrate assignment and potential confounders. Because total calories were held fixed, comparisons across the randomized carbohydrate levels can reasonably be interpreted as effects of the carbohydrate treatment for these mice.

- (b) Suppose x were instead measured as total carbohydrate intake, in grams, over the 28 days of the experiment. What would be the new regression line, including its intercept and slope?

Solution. Let $x_{\text{new}} = 28x_{\text{old}}$. The old intercept is

$$a = \bar{y} - b\bar{x} = 167.3 - 4.6(16.2) = 92.78.$$

Since $x_{\text{old}} = x_{\text{new}}/28$,

$$\hat{y} = 92.78 + 4.6 \frac{x_{\text{new}}}{28} = 92.78 + 0.1643x_{\text{new}}.$$

Therefore the new line is $\hat{y} = 92.78 + 0.1643x_{\text{new}}$. The intercept is unchanged, while the slope is divided by 28.

3. (**Lecture 5 Practice, Problem 2**) The table displays both potential outcomes only so the causal quantities can be checked; real data would reveal at most one potential outcome per unit.

Unit	$Y_i(0)$	$Y_i(1)$	D_i	Observed outcome (Y_i)
A	4	7	1	7
B	6	7	1	7
C	2	4	0	2
D	5	6	0	5

- (a) Calculate every individual treatment effect and the average treatment effect (ATE) across the four units.

Solution. The individual treatment effects are

$$\begin{aligned} Y_A(1) - Y_A(0) &= 7 - 4 = 3, & Y_B(1) - Y_B(0) &= 7 - 6 = 1, \\ Y_C(1) - Y_C(0) &= 4 - 2 = 2, & Y_D(1) - Y_D(0) &= 6 - 5 = 1 \end{aligned}$$

Therefore,

$$\text{ATE} = \frac{3 + 1 + 2 + 1}{4} = \frac{7}{4} = 1.75$$

- (b) Verify the observed-outcome identity

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

for one treated and one untreated unit. Identify which potential outcome would be missing for each unit in an actual study.

Solution. For treated unit A,

$$D_A Y_A(1) + (1 - D_A) Y_A(0) = 1(7) + 0(4) = 7$$

For untreated unit C,

$$D_C Y_C(1) + (1 - D_C) Y_C(0) = 0(4) + 1(2) = 2$$

Thus the observed outcomes are (7, 7, 2, 5). In an actual study, $Y_A(0)$ and $Y_B(0)$ would be missing for the treated units, while $Y_C(1)$ and $Y_D(1)$ would be missing for the untreated units.

- (c) Calculate the average treatment effect among the treated units.

Solution. The treated units are A and B, so their average treatment effect is

$$\frac{(7 - 4) + (7 - 6)}{2} = \frac{3 + 1}{2} = 2$$

- (d) Calculate the observed treated-minus-untreated mean difference.

Solution. The treated observed mean is 7, while the untreated observed mean is $(2 + 5)/2 = 3.5$. Therefore,

$$7 - 3.5 = 3.5$$

- (e) Calculate the untreated-potential-outcome selection term $\overline{Y(0)}_{D=1} - \overline{Y(0)}_{D=0}$, and verify that the average treatment effect among the treated plus this term equals the observed difference.

Solution. The treated units have mean untreated potential outcome $(4 + 6)/2 = 5$, while the untreated units have mean untreated potential outcome $(2 + 5)/2 = 3.5$. Thus the selection term is

$$5 - 3.5 = 1.5$$

The decomposition is

$$\underbrace{3.5}_{\text{observed difference}} = \underbrace{2}_{\text{average effect among treated}} + \underbrace{1.5}_{\text{selection term}}$$

- (f) Explain in words why the observed difference is not the ATE and why the counterfactual problem would prevent this decomposition from being calculated directly in real observational data.

Solution. The observed difference is not the ATE because it compares two different groups rather than the same units under treatment and no treatment. Here the treated units have an average treatment effect of 2, rather than the overall ATE of 1.75, and they would also have had higher average outcomes without treatment than the untreated units. The observed difference therefore combines a treatment effect with a selection difference. In real observational data, $Y_i(0)$ is missing for treated units and $Y_i(1)$ is missing for untreated units, so the two components cannot be separated without a research design or identifying assumptions.

4. The administration of Princeton University notices a strong positive correlation between students' academic achievement and whether they play an instrument. Admission Officer A hears of this finding and says, "Given that we want to admit students who do well academically, we should admit more students who play an instrument." Admission Officer B counters, "But this is just a correlation, and it does not imply that playing an instrument causes students to do well in their studies at Princeton University."

Who is right? Explain using relevant terminology, and relate your discussion to the specific issue of playing an instrument and academic success.

Solution. Both officers identify a valid point, although they are answering different questions. Officer A is making a *predictive* claim: if the association persists for future applicants, instrument-playing may contain information useful for predicting academic performance. Prediction does not require instrument-playing itself to cause the outcome.

Officer B is making a *causal* claim and is also correct. The observed association does not isolate the effect of playing an instrument. A common cause—for example, parental resources, prior preparation, or family emphasis on education—could affect both music participation and achievement. Reverse causality is also possible: students already doing well academically may have more time or opportunity for music. Without random assignment or another credible identification strategy, the correlation alone does not identify a causal effect.

4 Extra practice problems

1. Using similar techniques as for temperature, researchers reconstructed the average concentration of CO_2 in Earth's atmosphere over the last 410,000 years, measured in parts per million by volume. The variable x_i , $i = 1, \dots, 82$, contains the historic concentrations of CO_2 in 5,000-year intervals. The variable y_i records the difference between today's average global temperature and the historic global temperature, in degrees Celsius.

The means and standard deviations are

$$\bar{y} = -4.9, \quad \bar{x} = 227.3, \quad s_y = 2.64, \quad s_x = 28.3$$

The least-squares regression line for a regression of y on x is

$$\hat{y} = 0.081x - 23.31$$

- (a) What is your prediction of the historic relative temperature, in degrees Celsius, for a time when the CO_2 concentration was one standard deviation below its overall average?

Solution. One standard deviation below the mean is

$$x = \bar{x} - s_x = 227.3 - 28.3 = 199.0$$

Therefore,

$$\hat{y} = 0.081(199.0) - 23.31 = -7.191$$

The predicted relative temperature is approximately -7.19°C .

- (b) How confident are you about your prediction in part (a)? Compute a relevant measure and describe its interpretation for assessing predictions based on regression lines.

Solution. Since $b = r(s_y/s_x)$,

$$r = b \frac{s_x}{s_y} = 0.081 \frac{28.3}{2.64} \approx 0.868, \quad r^2 \approx 0.754$$

Thus approximately 75.4% of the sample variation in historic relative temperature is accounted for by its fitted linear relationship with CO_2 . This indicates a fairly strong linear fit within the observed data, but it does not make every individual prediction exact and does not by itself establish causality.

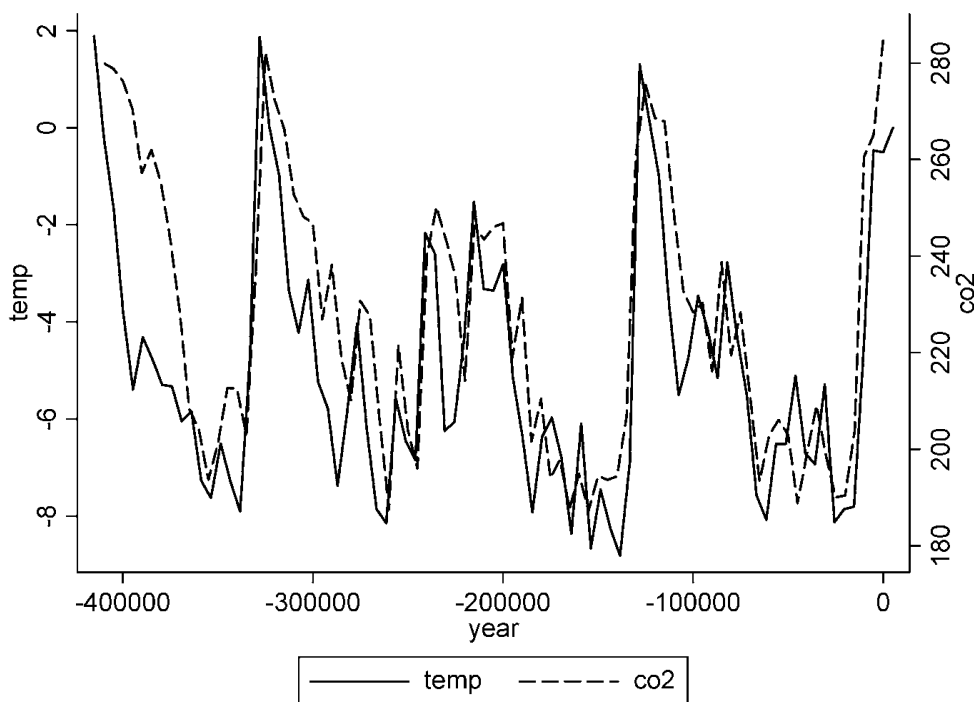
- (c) The CO₂ concentration was predicted to be 570 parts per million by volume in the year 2050. Compute the z -score of this value of x . Would you have much confidence in a prediction of the relative temperature in 2050 based on the regression line, given your result?

Solution. The standardized value is

$$z = \frac{570 - 227.3}{28.3} \approx 12.11$$

This value is more than 12 standard deviations above the historical mean and is far outside the range on which the line was fitted. A prediction there would be extreme extrapolation, so we should have little confidence in it, even though the in-sample value of r^2 is fairly high.

- (d) The figure below plots the joint evolution of relative temperature and CO₂ concentration over the last 410,000 years. Does the plot suggest that declines in CO₂ caused temperature to decline, or rather that declines in temperature caused CO₂ to decline? Why?



Historic relative temperature (solid) and CO₂ concentration (dashed).

Solution. The turning points in the plotted temperature series appear generally to precede the corresponding turning points in CO₂, so the graph is more consistent with temperature declines occurring first and CO₂ declines following. However, temporal ordering in an observational time series does not by itself prove either causal story: common causes, feedback, and other omitted forces may affect both series.

2. (a) Let $\hat{y}_i$, $i = 1, \dots, n$, be the predicted values from a regression of y_i on x_i . Suppose

$$\bar{x} = 3.4, \quad \bar{y} = -5.9, \quad \bar{\hat{y}} = -5.9, \quad s_x = 3.4, \quad s_y = 6.1, \quad s_{\hat{y}} = 2.2.$$

Say as much as you can about the correlation between x_i and y_i . Explain.

Solution. Because $s_{\hat{y}} = |r|s_y$,

$$|r| = \frac{2.2}{6.1} \approx 0.361.$$

Therefore $r \approx 0.361$ or $r \approx -0.361$. The information provided does not determine the sign. The averages do not provide additional information about the correlation.

- (b) Give an example of an association between two variables that is plausibly due to a common response.

Solution. For example, among cars on the road, braking distance may be positively associated with pollution. Vehicle age is a plausible common cause: older cars may both require longer braking distances and emit more pollution. The association between braking distance and pollution therefore need not reflect a causal effect of either variable on the other.

3. State whether each claim is true or false and explain briefly.

- (a) While the mean and standard deviation are not resistant measures, the regression slope is resistant.

Solution. False. The regression slope depends on the means, standard deviations, and correlation. A point with an extreme x -value can have high leverage and can substantially change the fitted slope.

- (b) If there is no linear association between two variables, then the correlation r is close to zero.

Solution. Under the usual definition, True: correlation measures the direction and strength of linear association. The important converse is false: r near zero does not imply that there is no association, because a strong nonlinear relationship can have correlation near zero. Outliers can also make r a misleading summary, so a scatterplot should be inspected.

- (c) The mean of the regression residuals is always zero.

Solution. True, for least squares with an intercept. Since $\hat{y}_i = a + bx_i$ and $a = \bar{y} - b\bar{x}$,

$$\bar{e} = \bar{y} - \bar{\hat{y}} = \bar{y} - (a + b\bar{x}) = 0.$$

- (d) If x and y are standardized, then the regression line from regressing y on x coincides with the regression line from regressing x on y .

Solution. True. A regression of y on x gives

$$a = \bar{y} - b\bar{x}, \quad b = r \frac{s_y}{s_x},$$

while a regression of x on y gives

$$\alpha = \bar{x} - \beta\bar{y}, \quad \beta = r \frac{s_x}{s_y}.$$

If x and y are standardized, then

$$s_x = s_y = 1 \quad \text{and} \quad \bar{x} = \bar{y} = 0.$$

Therefore,

$$\boxed{b = \beta = r \quad \text{and} \quad \alpha = a = 0}.$$

Thus, the two standardized regression lines coincide.

- (e) One explanation for an observed association between two variables is that one causes the other.

Solution. True. Direct causation is one possible explanation. Other possibilities include reverse causation, a common cause or confounder, selection, and chance. Association alone does not determine which explanation is correct.

4. (a) The height of men in the United States is approximately normally distributed with mean 68 inches and standard deviation 5 inches. Their weight has mean 168 pounds and standard deviation 25 pounds. The correlation between weight and height is 0.35. What is your best guess of a man's height if you are told that he weighs 160 pounds?

Solution. Let w denote weight and h height. The slope from regressing height on weight is

$$b = 0.35 \frac{5}{25} = 0.07,$$

and the intercept is

$$a = 68 - 0.07(168) = 56.24.$$

At $w = 160$,

$$\hat{h} = 56.24 + 0.07(160) = \boxed{67.44 \text{ inches}}.$$

- (b) Identify two mistakes in the following data description:

“The level of snow y_i , $i = 1, \dots, 365$, on Mount McKinley, measured daily in inches, has mean 83 inches, median 78 inches, interquartile range 42 inches, and variance 900 inches². The distribution is skewed to the right. Temperature x_i , in degrees Fahrenheit, has mean 18 and standard deviation 20. The least-squares regression line

$$\hat{y} = a + bx$$

has $a = 47$ degrees Fahrenheit and $b = 2$ inches per degree Fahrenheit.”

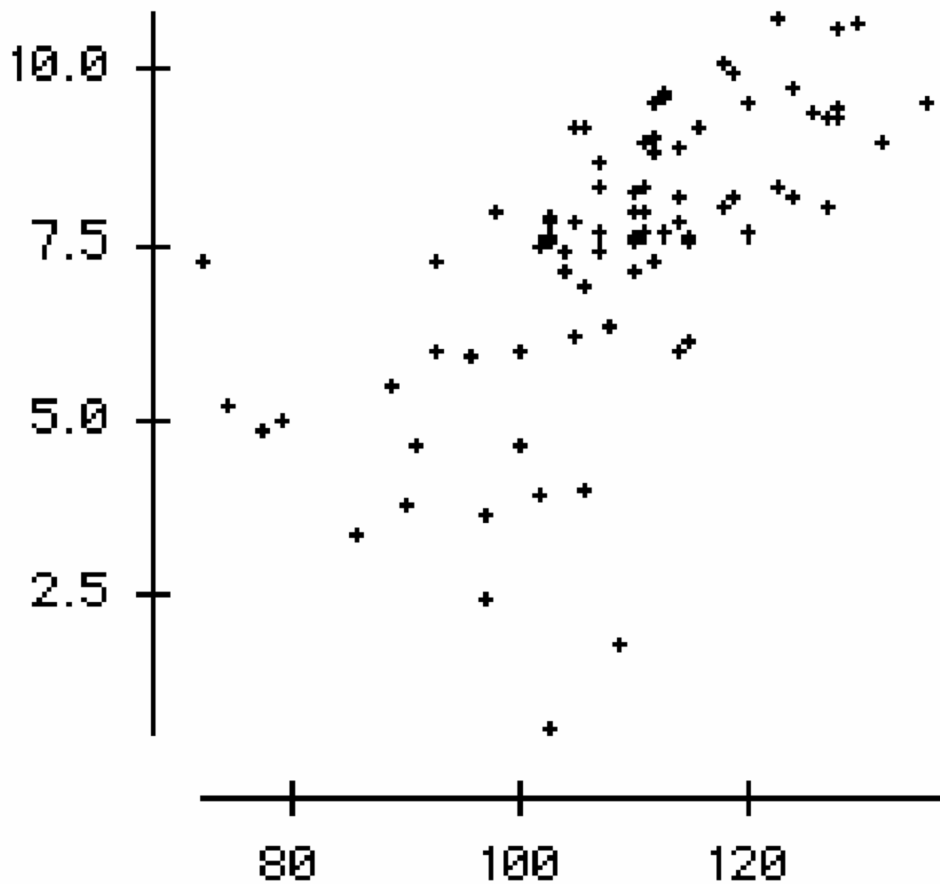
Solution. First, the intercept must have the units of the response variable, so a should be measured in *inches*, not degrees Fahrenheit.

Second, $s_y = \sqrt{900} = 30$ inches. The reported slope would imply

$$r = b \frac{s_x}{s_y} = 2 \frac{20}{30} = \frac{4}{3} > 1,$$

which is impossible. Therefore the reported summaries cannot all be correct.

- (c) True or false: the correlation of the data in the supplied scatterplot is closer to 0.2 than to 0.6.



Scatterplot supplied with Midterm I 2006, Question 4(b).

Solution. False. The points exhibit a moderately strong positive linear pattern, so 0.6 is more plausible than 0.2.

- (d) True or false: outliers of the explanatory variable are often influential in determining the least-squares regression line.

Solution. True. An observation far from $\bar{x}$ has high leverage. Because least squares and its component summaries are not resistant, such a point can substantially change the fitted slope and intercept, particularly when it also has a large residual.